

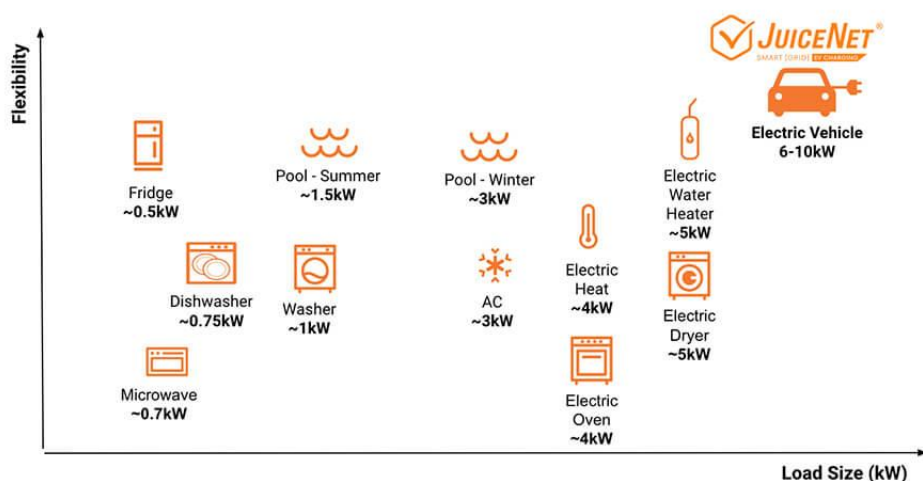
Topic 3. Incentivized or explicit flexibility

In this video we will analyse the conditions for consumers to provide explicit flexibility in return for incentives.

Consumers can provide flexibility if they have some equipment that produces or consumes a moderate or large amount of energy, such as:

- electrical heating or cooling systems such as air conditioning or heat pumps
- chargers of electric vehicles
- solar panels or domestic windmills
- sauna, pool purifier, automatic watering, and similar electrical-powered systems

If you have only a fridge or a microwave, then you may not be eligible: the technology necessary to connect you to the grid is much more expensive and environmentally impactful than the kilowatts that can be saved or flexibilised, as the figure shows.



Source: <https://www.enelxway.com/us/en/resources/blog/why-electric-vehicle-grid-integration-matters-today>

In addition to equipment, consumers need to be connected with other actors in the grid, namely producers, system and network operators or aggregators. Aggregators is a new actor that acts as a middleman between consumers and system operators: this actor “buys” flexibility services from consumers and “sells” them to producers and transporters when they need them.

To connect consumers with these actors, it is necessary to install smart meters and submeters in the equipment. These actors can then not only know if you are consuming or not, but also

can control the operations of your equipment remotely. Consumers are provided with an app where they can see the consumption of their equipment, set the times and days when they want to provide flexibility or decide with which equipment, they are willing to provide flexibility.

To recap, appropriate equipment and communication devices are necessary conditions to provide flexibility.

The provision of flexibility can be done in different ways.

One way is the so-called direct control. If you have agreed to provide flexibility to an aggregator in a specific event in the future, the aggregator will send you a message to inform you that the event is approaching and when the day/time comes this agent will remotely control your equipment. This may imply using more or using less energy.

Alternatively, consumers can commit to participate in these “events” with no direct control. Consumers are sent an alert by SMS or email and if they comply with the signal, they receive compensation.

Direct control works because consumers have different practices and comfort thresholds. Maybe you cannot do without air conditioning or heating for an hour, but there are other consumers that do not mind. And reversely, maybe these consumers cannot wait two hours to use the oven, but you do not mind. The diversity in consumer profiles enables the success of direct control. Notice that direct control services are very different in duration (from 15 minutes to two hours), anticipation (a week's notice or a year's notice) and in the equipment that you commit (the air conditioning or the charger of the EV). These are only examples to show that direct control can be implemented in many ways depending on the local context.

Another way of explicit flexibility is flexibility by design or automatized flexibility. This is what smart EV chargers do. You install the EV charger and set the conditions you want to obtain. Say, to have the vehicle charged at 80% by 8.30 am. Then, the device starts and stops charging depending on the prices. This is done automatically, so the consumer does not have to carry the burden of being attentive to prices.

With these smart chargers, EVs can also discharge the battery to give power when necessary. If the car is fully charged but you are not using it, the energy stored can be fed into the system if needed. And conversely, if there is plenty of solar energy, they will charge your car, so that no excess energy is created in the grid.

These smart systems can be used for flexibility provision for other equipment. You can install a smart system for your heating-cooling. You tell the system when you need your house warmed and at which temperature. The system will turn on and off the heating/cooling

considering manifold points of information such as hourly tariffs, carbon emissions, external temperature, home insulation, home size, etc. The thermal storage will maintain an adequate temperature when you need it.

Flexibility by design creates savings for consumers: as their equipment is on only when they really need it, they save energy and money. The energy that is not wasted can be then aggregated and sold to producers if they need it.

In sum, direct control and flexibility by design are ways of providing flexibility to the system. These systems show that consumers can compensate for the limitations of intermittent systems and ensure that we reap the tremendous benefits of having a decarbonized system. We will discuss the benefits for consumers in the next video.